01 Serve, stage by stage

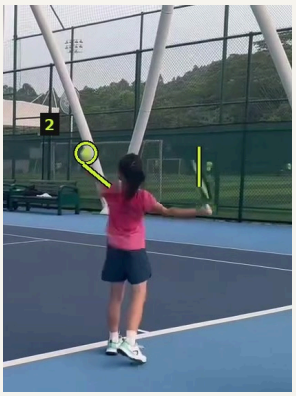
Serve 1 of 2 in the video, split into the 8 stages coaches and sport scientists use for the serve (Kovacs & Ellenbecker). Serve 2 shows the same pattern.



1 · 9.30 s

Start

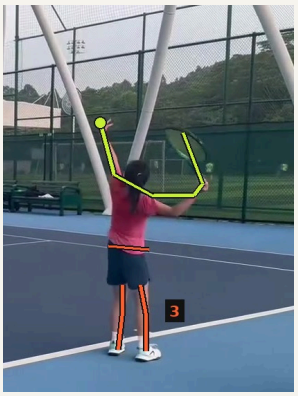
1 Feet close, line of the feet almost parallel to the baseline.



2 · 9.83 s

Release

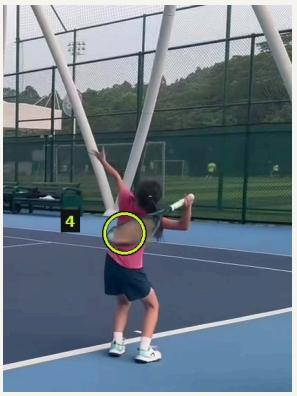
2 Ball leaves the hand at head height, racket goes up with it.



3 · 10.03 s

Loading · trophy

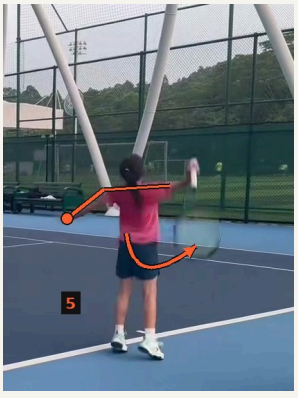
3 Arms good, but the racket face is wide open. Hips square, knees straight.



4 · 10.45 s

Cocking · racket drop

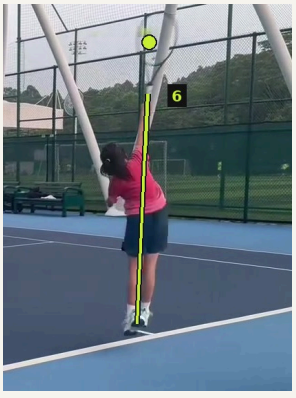
4 Racket drops behind the back.



5 · 10.93 s

Acceleration

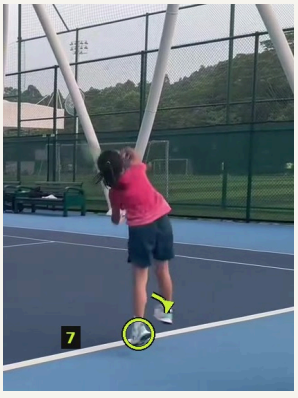
5 Racket comes out to the side, left arm opens. Loop.



6 · 11.10 s

Contact

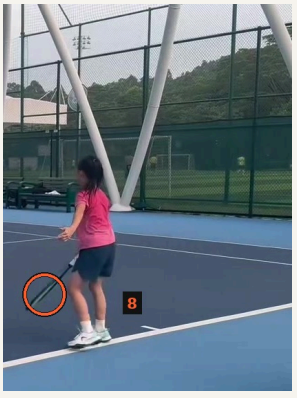
6 Full reach, but she hits with a flat face, like a frying pan.



7 · 11.23 s

Deceleration · landing

7 Lands on the left foot just inside the baseline.



8 · 12.40 s

Finish · recovery

8 Walks in with the racket low, no ready position.

In one look

A good base: high toss arm, a real racket drop, full reach and a balanced landing. The main problem is the grip: the racket face is open at the trophy and she hits the ball flat, like a frying pan (stages 3 and 6). On top of that the body is not loaded at the trophy, and the racket loops out to the side before it goes up (stage 5).

Grip first. With a continental grip the loop and the frying pan usually get smaller on their own.

P.K.

02 What costs speed

Trophy at 10.03 s, and the same moment before the swing up in both serves.

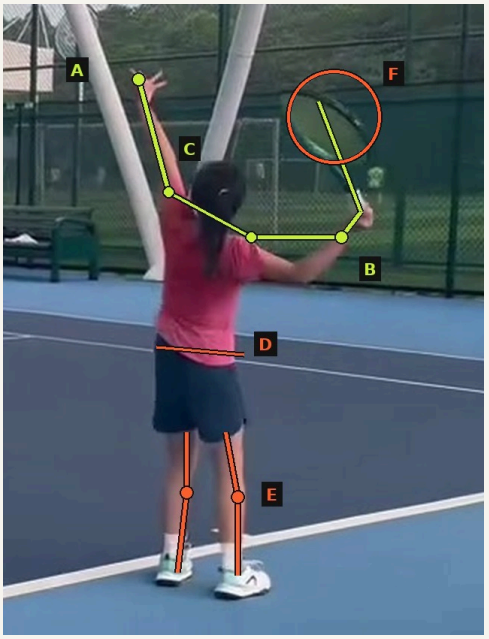


Fig. 1 · Trophy, serve 1, 10.03 s

Trophy

CHECK	HERE	REFERENCE
A Tossing arm	Long, straight	Fully extended
B Racket arm	Elbow at shoulder height, tip up	Elbow level with shoulders
C Shoulder tilt	Front shoulder higher, about 25–30° in this view	About 40–45°
D Hips, shoulder turn	Almost square to the net	Turned, back towards the target
E Knee bend	Legs almost straight	Clear bend; studies report about 47–81° by age
F Racket face	Wide open to the sky, frying-pan position	Edge-on, continental grip

Grip. An open face at the trophy is a sign of a forehand-type grip instead of continental. With it the racket cannot come edge-first to the ball and the forearm cannot turn over (pronate) into contact, so she pushes the ball flat: little spin, little control, and it limits speed.

Research links more knee bend at the trophy to higher racket speed. Angles here are read from a 2D view from behind, so treat them as estimates.

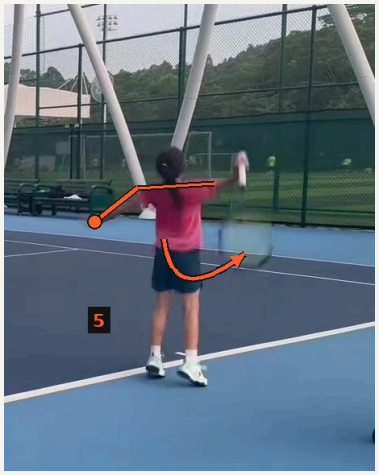


Fig. 2 · Serve 1, 10.93 s

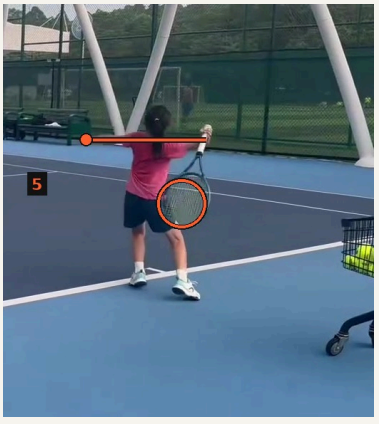


Fig. 3 · Serve 2, 36.55 s

The loop

After the drop behind the back, the racket comes out to the side with the arms in a T, and only then goes up. It happens in both serves, so it is a habit, not a one-off.

From the racket reaching the back to contact: about `0.7–0.8 s` in both serves. The swing waits instead of flowing.

03 Checklist and plan

Each stage against the reference model.

STAGE	WHAT WE SEE	REFERENCE	VERDICT
1 Start	Feet close, nearly parallel to the baseline	Platform: feet about shoulder-width apart	Work on
2 Release	Ball released at head height, racket up together	Straight arm, smooth lift	Good
3 Loading	Arms good; racket face open; hips square, knees almost straight	Turn, shoulder tilt, clear knee bend, racket edge-on	Work on
4 Cocking	Real drop behind the back	Racket drops behind the back	Good
5 Acceleration	Loop to the side, left arm opens early, ~0.7–0.8 s wait	Continuous drop and swing up, left arm tucks	Work on
6 Contact	Full reach, jumps up, but flat face at contact (frying pan)	Arm extended, forearm turns over into the ball	Work on
7 Landing	Left foot, just inside the baseline, back leg through	Front foot inside the court, balanced	Good
8 Recovery	Walks in, racket low, no ready position	Balance, then ready for the return	Work on
Grip	Coach: forehand-type grip, face open at trophy and contact	Continental grip	Work on
Toss in front	Not visible from behind	Needs a side view	Side view

Priorities

1. **Continental grip.** Check it before every serve until it is automatic.
2. **One continuous swing.** From the drop straight up to the ball, no stop at the side.
3. **Load the trophy.** Turn the back to the target and bend the knees while the ball goes up.
4. **Ready after landing.** Two steps to balance, split step, racket in front.

Drills, two weeks

Edge first. Continental grip, swing with the edge to the ball, start with slice serves.	3×10
Start from the drop. Racket already behind the back, toss, swing up.	3×10
Towel or ball throw. Overhead throw over the net, same motion, no stop.	2×10
Trophy freeze. Toss, hold the trophy turned and bent, catch the ball.	10
Serve and split. Serve, land, split step on the coach's clap.	10

References. Kovacs & Ellenbecker (2011), 8-stage serve model, Sports Health. Jacquier-Bret & Gorce (2024), serve kinematics review, Frontiers in Sports and Active Living. Mouratoglou Academy, trophy position. Martin (2015), serve stance, ITF Coaching & Sport Science Review.